MATH 120 Exam II

① $A = \begin{bmatrix} 1 & -2 & 3 & 1 \\ 5 & -9 & 6 & 3 \\ -1 & 2 & -6 & -2 \\ 2 & 7 & 6 & 3 \end{bmatrix} \xrightarrow[R3=R3+R1]{R2=R2+5R3,\ R4=R4+2R3} \begin{bmatrix} 1 & -2 & 3 & 1 \\ 0 & 1 & -24 & -7 \\ 0 & 0 & -3 & -1 \\ 0 & 11 & -6 & -1 \end{bmatrix} \xrightarrow{R4=R4-11R2}$

$\begin{bmatrix} 1 & -2 & 3 & 1 \\ 0 & 1 & -24 & -7 \\ 0 & 0 & -3 & -1 \\ 0 & 0 & 258 & 76 \end{bmatrix} \xrightarrow{R4=R4+86R3} \begin{bmatrix} 1 & -2 & 3 & 1 \\ 0 & 1 & -24 & -7 \\ 0 & 0 & -3 & -1 \\ 0 & 0 & 0 & -10 \end{bmatrix}$

$\boxed{\det(A) = 1 \times 1 \times -3 \times -10 = 30}$

② $A^3 = A$

$A^3 - A = 0$

$A(A^2 - c) = 0$

$\det(A)\det(A^2 - c) = \det(0)$

↳ $\det(A) = 0$

↳ $\det(A^2 - c) = 0$

$A^2 - c = 0$

$A^{-1}A^2 - A^{-1}c = 0$

$A = A^{-1}$

$\det(A) = \det(A^{-1})$

$[\det(A)]^2 = 1$

$\det(A) = \pm 1$

$\boxed{\text{so } \det(A) \text{ has 3 possible values: } +1,\ 0,\ \& -1}$

③

1. $0^2 + 0^2 + 0^2 = 0 \leq 1$

it does have a zero vector

2. ~~$u = (1,0,0) = 1 \leq 1$~~

~~$2u = (2,0,0) =$~~

$u = (1,0,0)$

$\|u\|^2 \leq 1$

$2u = (2,0,0)$

$\|2u\|^2 = 4 \not\leq 1$

it does not fit all properties of subspace already, therefore $\boxed{\text{No}}$

④ $A = 4$ & $A(5,7)$

Dim Row A = Rank A = 4

Dim Col A = Rank A = 4

Dim Nul A = #Col − Rank A = 3

$A^T = 4$ & $A^T(7,5)$

Dim Row A^T = Rank A^T = 4

Dim Col A^T = Rank A^T = 4

Dim Nul A^T = #Col A^T − Rank A^T = 1

⑤ $B = \left\{ \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 5 \\ -6 \end{bmatrix} \right\}$, $x = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$

$$c_1 \begin{bmatrix} 1 \\ -2 \end{bmatrix} + c_2 \begin{bmatrix} 5 \\ -6 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$$

$c_1 + 5c_2 = 4$ ①

$-2c_1 - 6c_2 = 0$ ②

① $c_1 + 5c_2 = 4$

$c_1 = 4 - 5c_2$ ③

③ → ② $-8 + 10c_2 - 6c_2 = 0$

$4c_2 = 8$

$\boxed{c_2 = 2}$ ④

④ → ① $c_1 + 10 = 4$

$\boxed{c_1 = -6}$

$$[x]_B = \begin{bmatrix} -6 \\ 2 \end{bmatrix}$$

(B) since $v_1, v_2, \cdots, v_p$ are linearly dependent, there must be at least one non-zero scalar among $c_1, c_2, \cdots, c_p$:

$$c_1 v_1 + c_2 v_2 + \cdots + c_p v_p = 0$$

apply ~~trans~~ transformation by matrix A:

$$A(c_1 v_1 + c_2 v_2 + \cdots + c_p v_p) = A(0)$$

$$c_1 A v_1 + c_2 A v_2 + \cdots + c_p A v_p = 0$$

we ~~kn~~ know that at least one scalar among $c_1, c_2, \cdots, c_p$ is non-zero, therefore,

$$A v_1 + A v_2 + \cdots + A v_p = 0$$

& $A v_1, A v_2, \cdots, A v_p$ are also linearly dependent.